RVHS JC2 H2 Physics Preliminary Examinations Paper 2 Solutions

1 (a) $S_y = u_y t + ½ at^2$
$20 = 0 + ½ (9.81)t^2$
$t = 2.02$ s

(b) No. As the time duration is larger than human reaction time, which is about 0.3 s.
or
No. As the rock reaches the intended target in more than 2.0 s time, the soldier would have moved away sufficiently far, assuming a speed of 1 m s⁻1, i.e. 2 m away.
or
No. Air resistance will render the horizontal distance travelled to be lesser than 50 m.

(c) $S_x = u_x t$
$50 = u_x(2.02) \rightarrow u_x = v_x = 24.752$
$v_y = u_x + (9.81)(2.02) \rightarrow \quad v_y = 19.816$
$v = (v_x^2 + v_x^2)^{½} = 31.7$ m s^{-1} ~ 32 m s^{-1}
$\tan\theta = 19.816/24.752$
$\theta = 38.7° \sim 39°$
39° below the horizontal

2 (a) <u>The linear momentum of a body is the product of the mass and its velocity.</u> The